

SEARCHING

PROBLEM 7

QUESTION:

DS	Session	Searching	Question Information	Level 1	Challenge 7
Question Description: Suresh have "N" rectangles. A rectangle is Silver if the ratio of its sides is in between $[1.6, 1.7]$, both inclusive. Your task is to find the number of silver rectangles. Constraints: $1 \leq N \leq 10^5$ $1 \leq W,$ $H \leq 10^9$ Input Format: First line: Integer "N" denoting the number of rectangles Each of the "N" following lines: Two integers W, H denoting the width and height of a rectangle Output Format: Print the output in a single line contains find the number of Silver rectangles. Sample Input: 5 10 1 165 100					

180 100
170 100
160 100
Sample Output:
3
Explanation:
There are three Silver rectangles: (165, 100), (170, 100), (160, 100).

Logical Test Cases

Test Case 1
INPUT (STDIN)
4
185 100
180 100
170 100
160 100
EXPECTED OUTPUT
2

Test Case 2
INPUT (STDIN)
6
185 100
180 100
170 100
160 100
165 90
164 95
EXPECTED OUTPUT
2

Mandatory Test Cases

Test Case 1

Test Case 2

KEYWORD	KEYWORD	
if(width/height>=1.6 && width/height<=1.7)	else if(height/width >=1.6 && height/width<=1.7)	
Complexity Test Cases		
Test Case 1	Test Case 2	Test Case 3
CYCLOMATIC COMPLEXITY	TOKEN COUNT	NLOC
7	125	25

SOURCE CODE:

```
#include <stdio.h>

int is_silver_ratio(double dim1, double dim2) {
    double r1 = dim1 / dim2;
    double r2 = dim2 / dim1;
    return ((r1 >= 1.6 && r1 <= 1.7) || (r2 >= 1.6 && r2 <= 1.7));
}

int main() {
    int total_rectangles;
    if (scanf("%d", &total_rectangles) != 1) return 0;

    int matching_count = 0;
    for (int i = 0; i < total_rectangles; i++) {
        double side_a, side_b;
        scanf("%lf %lf", &side_a, &side_b);
        if (is_silver_ratio(side_a, side_b)) {
            matching_count++;
        }
    }
    printf("%d\n", matching_count);
    return 0;
}
```

OUTPUT:

```
5
185 100
180 100
170 100
160 100
165 90
2
```

```
6
185 100
180 100
170 100
160 100
165 90
164 95
2
```